

hind the aircraft. Present-day missiles using advanced IR sensors can even lock on to friction-heated air streaming back from the aircraft's nose. These missiles can be fired from any angle.

IR-guided-missile seekers

Two broad categories of seekers are used in heat-seeking missiles. These are reticle seekers using single IR detectors and imaging seekers that use a focal plane array of IR/UV detectors. Both types have different variants.

In the most basic reticle seeker (Fig. 6), IR energy emitted by the target is collected by the optical system

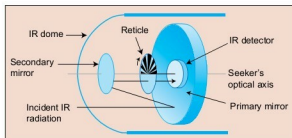


Fig. 6: Basic reticle seeker



Fig. 7: Representative reticles

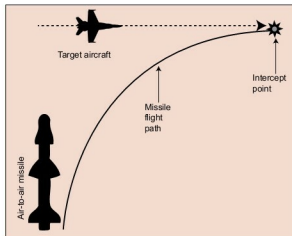


Fig. 8: Pure pursuit navigation

and focused on the detector through a rotating reticle. A reticle is nothing but an optical modulator made up of a circular element having sequentially-arranged transparent and opaque spokes on it (Fig. 7). The reticle chops the scene represented by IR energy. The output of the detector is a sequence of pulses whose amplitude is proportional to the magnitude of IR energy and whose frequency is equal to the product of spin rate of the reticle and the number of transparent/opaque spoke pairs.

Also, modern reticle-seeker based heat-seeking missiles use spinning mirrors and a fixed reticle. The phase of detector signal with respect to a reference phase is used to determine the angular position of the seeker axis with respect to the target. The angular error is then used to keep the seeker axis pointed towards the target within its field-of-view and

also control the flight path of the missile to intercept the target.

We have spin-scan seekers that suffer from the problem of centre-null and conical-scan seekers that eliminate this problem. We shall not go into details of these techniques. Lead-sulphide (PbS) with peak sensitivity in the wavelength region of two micron when uncooled, indium-

antimonide (InSb) with peak sensitivity in 4-5 micron band when cooled to liquid nitrogen temperature and mercury-cadmium-telluride (HgCdTe) with peak sensitivity in 8-12 micron band when cooled to liquid nitrogen temperature are the commonly used detector materials.

Lead sulphide detectors were used in older missile seekers, which forced missiles to look at stern engagements

as the missile had to look at the hot turbine in the tail-pipe region to get sufficient signal to track the target. Modern missiles almost invariably use InSb or HgCdTe detectors.

Missile seekers would find it very easy and convenient to detect and track a hot target against a uniform benign cool background than doing so in the presence of clouds and extended IR sources. Sunlight reflected from the edge of a cloud would be as attractive a target, if not more, for the missile seeker than a jet aircraft. Reticle seekers handle all these issues by having very small instantaneous field-of-view (IFOV) and/or advanced signal processing.

Navigation to the target is the next important step after target acquisition and tracking. In one method called pure pursuit, also called direct pursuit, the navigation technique ensures that the seeker is looking at the target continuously throughout the engagement duration till it eventually hits the target. In the case of pure pursuit engagement, the flight path is not the most direct one, as shown in Fig. 8, and the trajectory has an ever-decreasing radius-turn towards the end of the engagement. This poses a problem as the missile may not be left with enough energy to complete the turn in close-in range, allowing the target to escape.

Proportional navigation or proportional pursuit, which is invariably followed in air-to-air missiles, allows the missile to have the shortest flight path towards the intercept, and therefore eliminates the need for a high-g manoeuvre towards the end. Proportional navigation flight path is established by a constant look angle as shown in Fig. 9 for a given constant missile velocity and assuming that target does not manoeuvre. In practice, target does manoeuvre and the missile velocity is also not constant through engagement duration. Therefore look angle is updated as and when required.

A variant of the reticle seeker is the pseudo imaging seeker. This seeker uses one or more detectors enabling both spatial and temporal infor-